

AP Final Project

University ERP (Java + Swing)

Group Members

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How To Run

***This project was made with Maven to handle dependencies**

***Run on Windows Powershell**

Prerequisites:

Java Version "21.0.7"

java version "21.0.7" 2025-04-15 LTS

Java(TM) SE Runtime Environment (build 21.0.7+8-LTS-245)

Java HotSpot(TM) 64-Bit Server VM (build 21.0.7+8-LTS-245, mixed mode, sharing)

Maven Version 3.9.11

Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)

Maven home: C:\Program Files\apache-maven-3.9.11

Java version: 24, vendor: Oracle Corporation, runtime:

C:\Users\Mayank\OneDrive\Desktop\java\OpenJDK-24

Default locale: en_IN, platform encoding: UTF-8

OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"

Windows Powershell

MySQL

Database Configuration

Download the files.

Run the seed scripts on MySQL. (To be taken from test pack)

Edit the DataSourceProvider in edu.univ.erp.util to add your username and password for both auth_db and erp_db database configuration.

Eg:

```
HikariConfig cfg = new HikariConfig();
cfg.setJdbcUrl("jdbc:mysql://localhost:3306/auth_db?useSSL=false&serverTimezone=UTC");
cfg.setUsername("//your username");
cfg.setPassword("//your password");
cfg.setMaximumPoolSize(10);
cfg.setMinimumIdle(2);
cfg.setConnectionTimeout(20000);
cfg.setPoolName("AuthHikariPool");
auth_ds = new HikariDataSource(cfg);
```

Compilation & Running*

mvn clean compile exec:java "-Dexec.mainClass=edu.univ.erp.Main"

Databases & Tables

AUTH_DB

1. AUTH_USERS table stores usernames, userid, password hash, names and login related metadata(like last login, failed attempts, locked until etc.)

ERP_DB

1. SETTINGS table stores maintenance mode on or off
2. STUDENTS table stores user_id, roll numbers, year and program
3. INSTRUCTORS table stores user_id, department and designation
4. SCORES table stores scores for assessments of all students using user_id and assessment_id
5. ASSESSMENTS table stores assessments for all sections using section_id, and stores max_marks and weight
6. FINAL_GRADES table stores final grades for all students for courses where grade is finalised
7. ADD_DROP_DEADLINE table stores add drop period deadline
8. ENROLLMENTS table stores enrollments of students in section
9. COURSES table stores all courses with department and credits

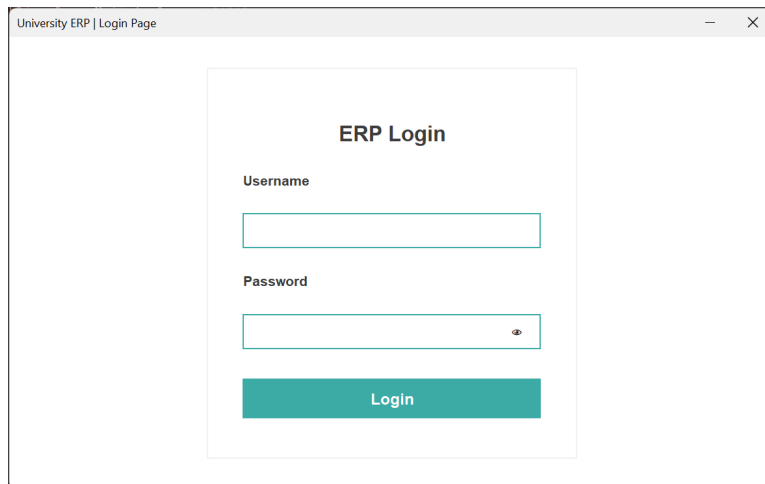
10. SECTIONS table stores all sections with instructor_id, day/time, room, course etc.

11. GRADE_SLABS table stores section wise grading slabs for final grade computation with min and max percentage for a letter grade

*For the exact structure of tables, seed commands in the test pack can be referred to.

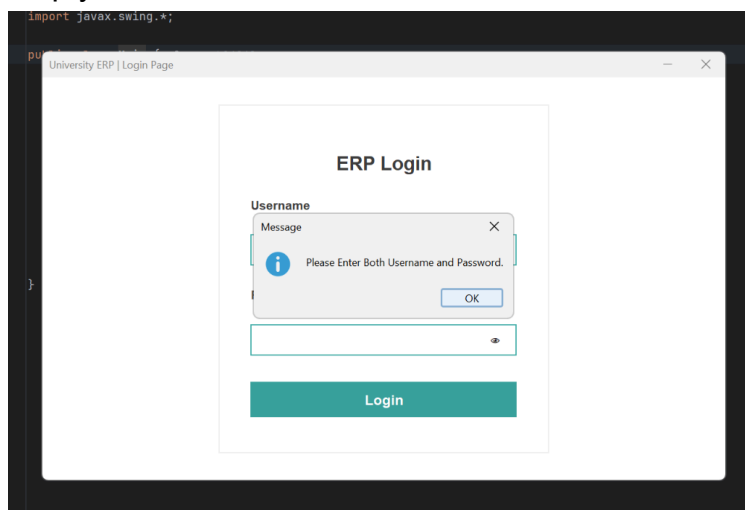
Screen Shots

Login screen

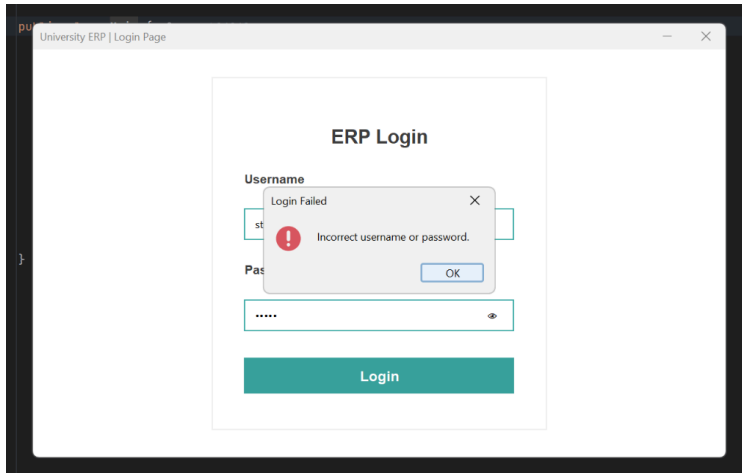


Login Screen Functionalities

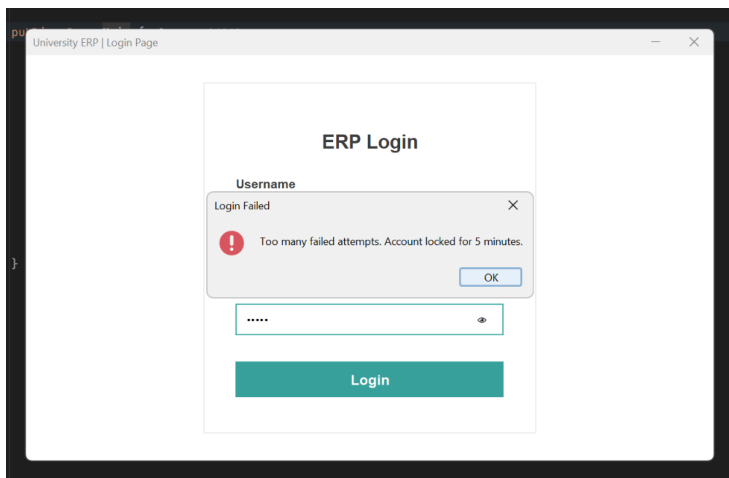
Empty Field Entered



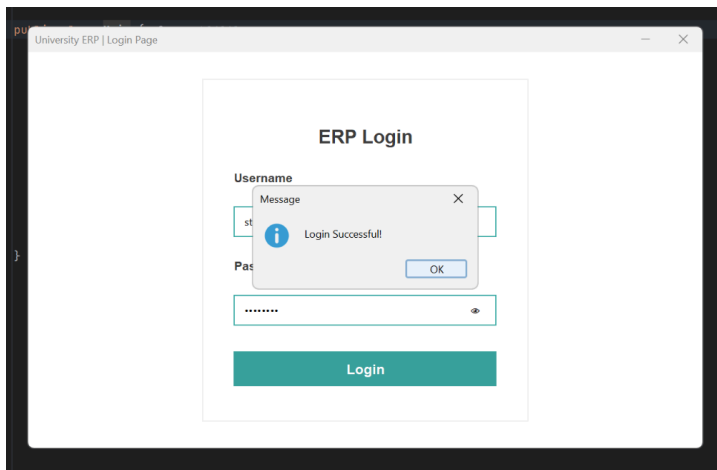
Incorrect username/password



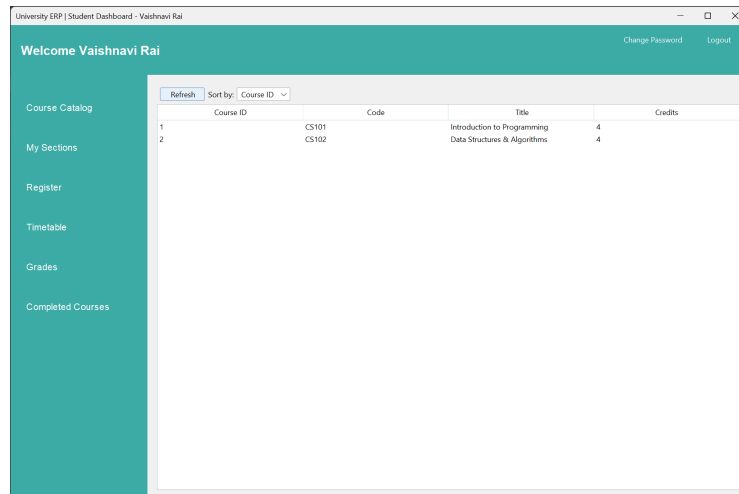
Logout after 5 fails



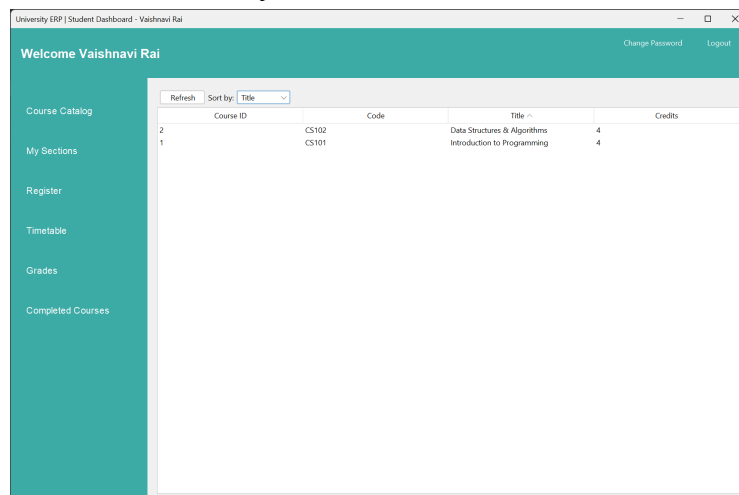
Successful login



Student Dashboard with Course Catalog (default sorted by Course ID)



Courses Sorted By Title



Password Change Dialog

?

Old Password:

.....

New Password:

.....|

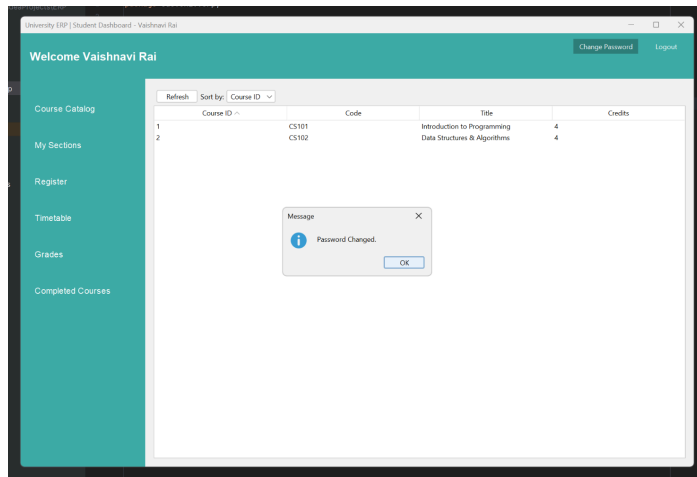
OK

Cancel

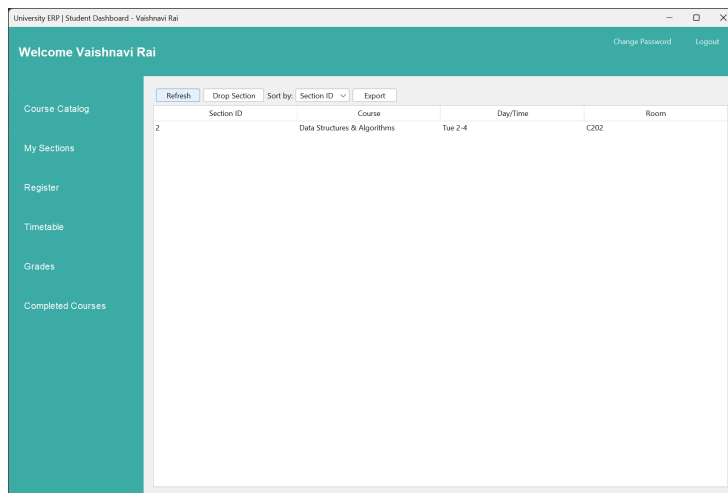
i

Old And New Password Cannot Be Same.

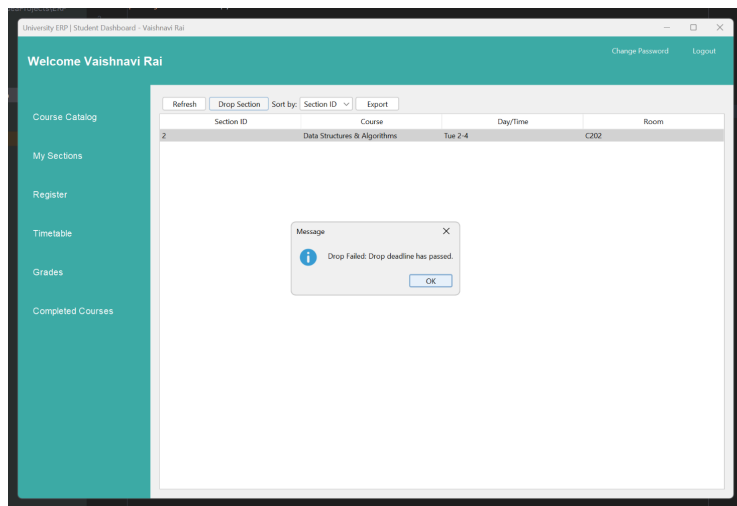
OK



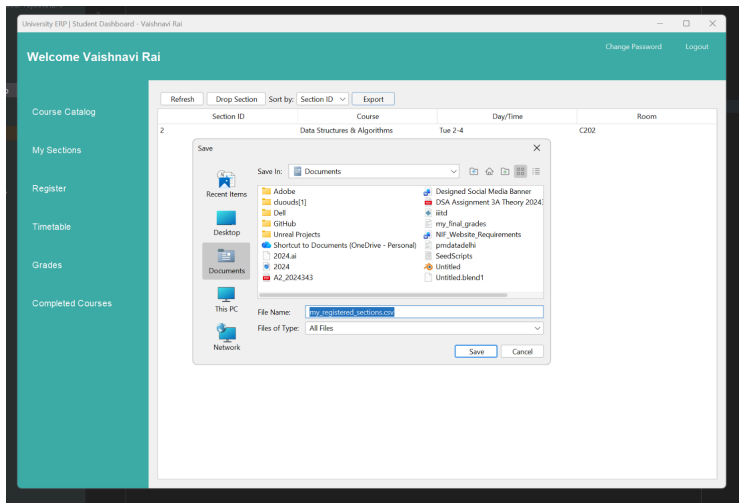
Registered Sections



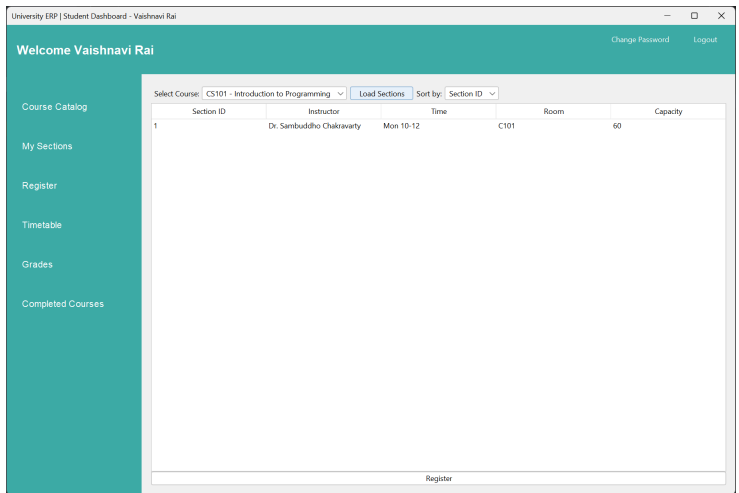
Drop after deadline fails



CSV Export



Course Registration



Grades Table & Export

University ERP | Student Dashboard - Vaishnavi Rai

Change Password | Logout

Welcome Vaishnavi Rai

Course Catalog

My Sections

Register

Timetable

Grades

Completed Courses

Load Final Grades | Load Breakdown | Export Final Grades

Section	Percentage	Grade
Introduction to Programming	88.0	A

Section	Assessment	Score	Final %
Data Structures & Algorithms	Quiz	14.0	82.0
Data Structures & Algorithms	Midterm	26.0	82.0
Data Structures & Algorithms	Endsem	42.0	82.0

University ERP | Student Dashboard - Vaishnavi Rai

Change Password | Logout

Welcome Vaishnavi Rai

Course Catalog

My Sections

Register

Timetable

Grades

Completed Courses

Load Final Grades | Load Breakdown | Export Final Grades

Section	Percentage	Grade
Introduction to Programming	88.0	A

Save

Save In: Documents

Recent Items

Desktop

Documents

This PC

Network

File Name

Files of Type

Save

Cancel

Section	Assessment	Score	Final %
Data Structures & Algorithms	Quiz	14.0	82.0
Data Structures & Algorithms	Midterm	26.0	82.0
Data Structures & Algorithms	Endsem	42.0	82.0

Finished Courses

University ERP | Student Dashboard - Vaishnavi Rai

Change Password | Logout

Welcome Vaishnavi Rai

Course Catalog

My Sections

Register

Timetable

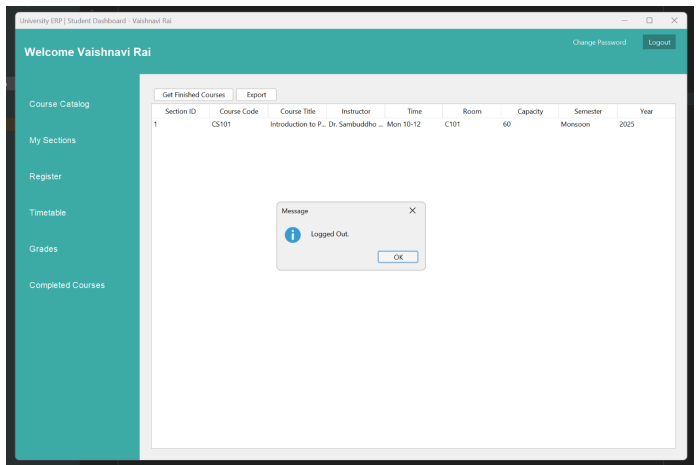
Grades

Completed Courses

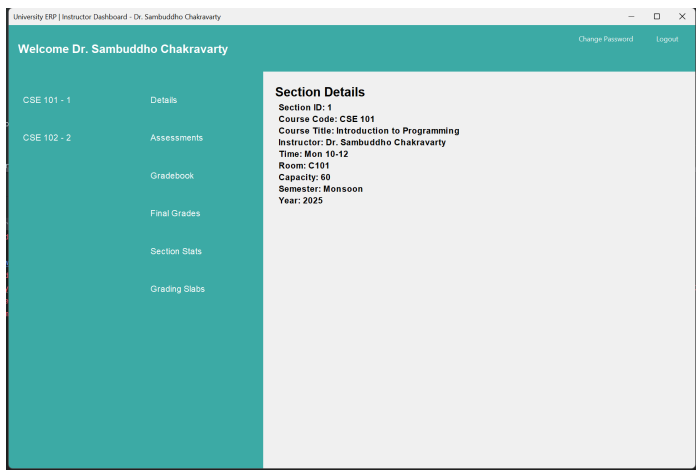
Get Finished Courses | Export

Section ID	Course Code	Course Title	Instructor	Time	Room	Capacity	Semester	Year
1	CS101	Introduction to P...	Dr. Sambuddho ..	Mon 10-12	C101	60	Monsoon	2025

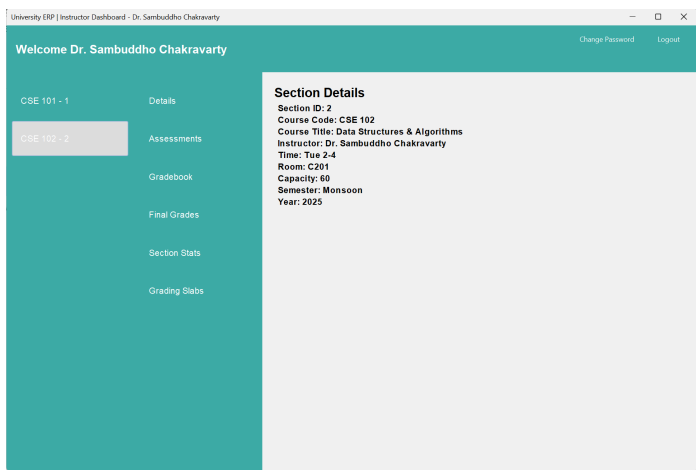
Logout



Instructor Dashboard



Section Selector



Assessments tab for section with add assessment, edit assessment and delete assessment

University ERP | Instructor Dashboard - Dr. Sambuddho Chakravarty

Welcome Dr. Sambuddho Chakravarty

CSE 101 - 1

CSE 102 - 2

Details

Assessments

Gradebook

Final Grades

Section Stats

Grading Slabs

LoadAddEditDelete

ID	Name	Max	Weight
1	Quiz	20.0	20.0
2	Midterm	30.0	20.0
7	Project	100.0	10.0
8	Endsem	50.0	50.0

University ERP | Instructor Dashboard - Dr. Sambuddho Chakravarty

Welcome Dr. Sambuddho Chakravarty

CSE 101 - 1

CSE 102 - 2

Details

Assessments

Gradebook

Final Grades

Section Stats

Grading Slabs

LoadAddEditDelete

ID	Name	Max	Weight
1	Quiz	20.0	20.0
2	Midterm	30.0	20.0
7	Project	100.0	10.0
8	Endsem	50.0	50.0

Add Assessment

Name:

Max Marks:

Weight (%):

OK

Cancel

Final Grades preview

University ERP | Instructor Dashboard - Dr. Sambuddho Chakravarty

Welcome Dr. Sambuddho Chakravarty

CSE 101 - 1

CSE 102 - 2

Details

Assessments

Gradebook

Final Grades

Section Stats

Grading Slabs

Sort by: Student

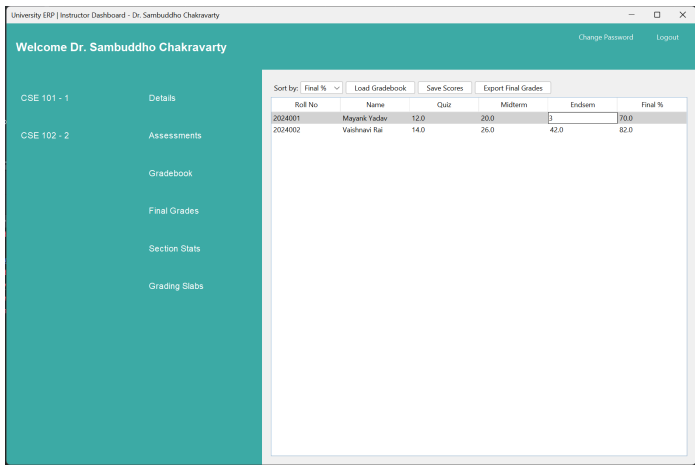
Preview Finals

Finalize Grades

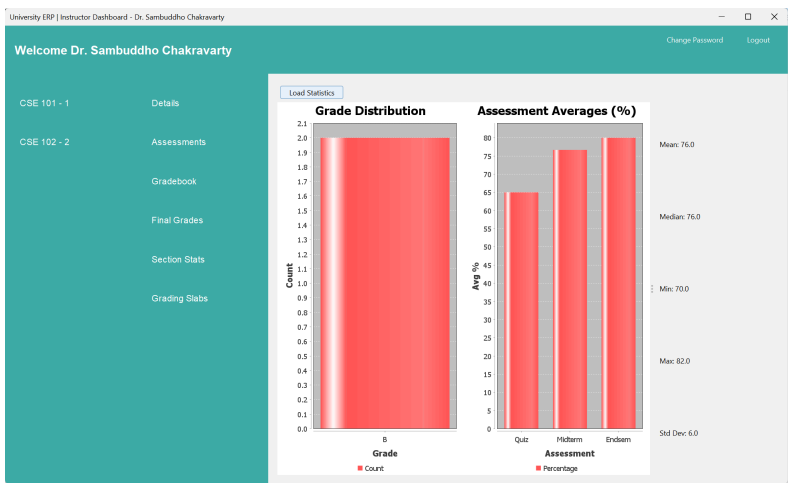
Export Final Grades

Student	Roll Number	Percentage	Grade
Mayank Yadav	2024001	70.0	B
Vaishnavi Rai	2024002	82.0	B

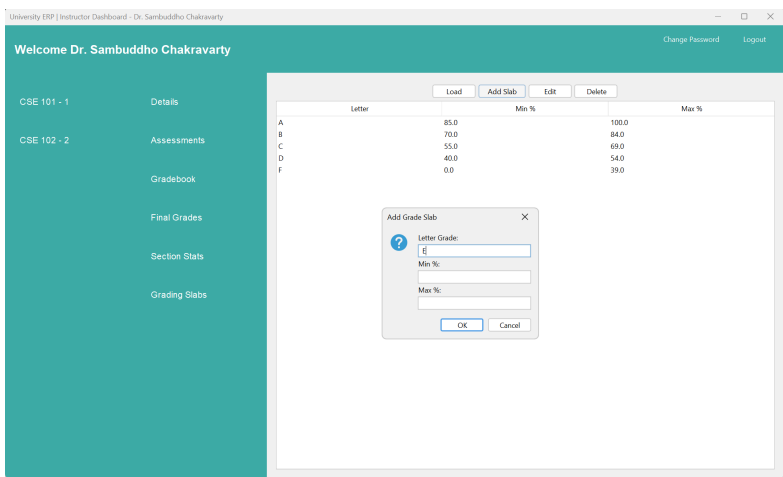
GradeBook & Score Entering



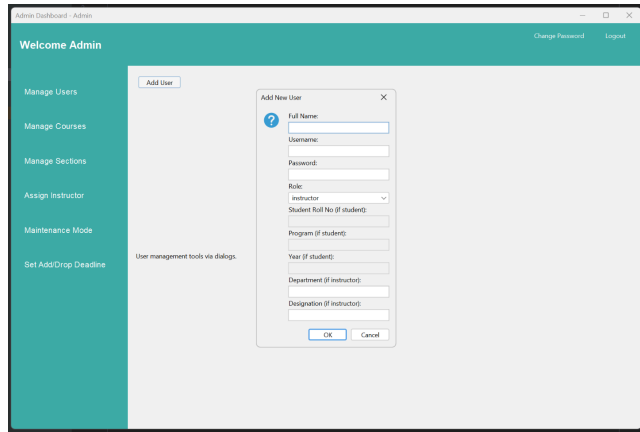
Section Stats



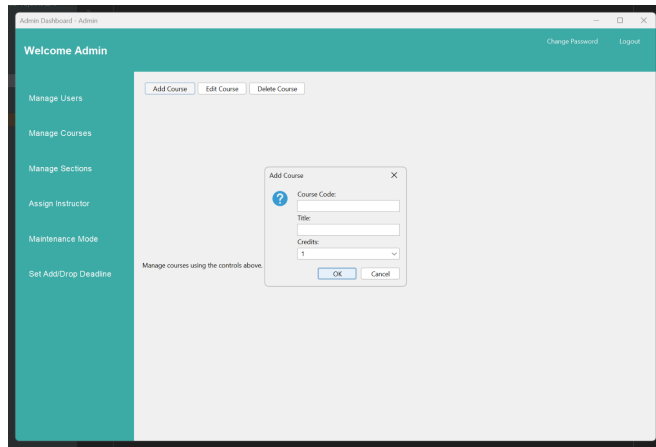
Slabs Panel



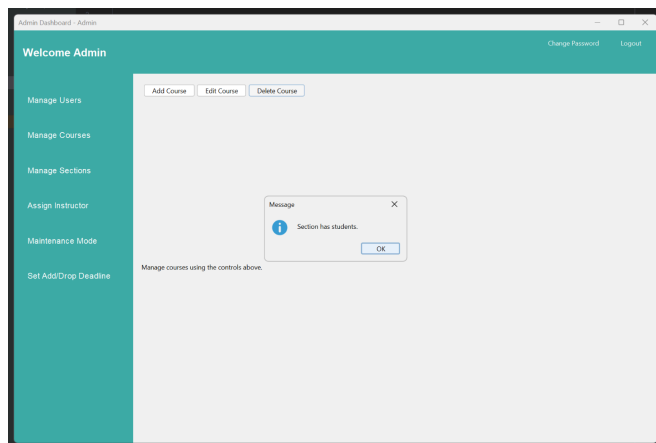
Admin Dashboard and Add User Dialog for Instructor



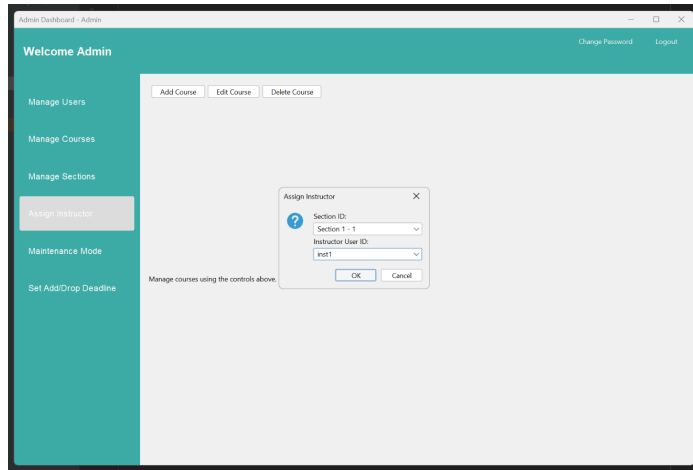
Courses Management & Add Course



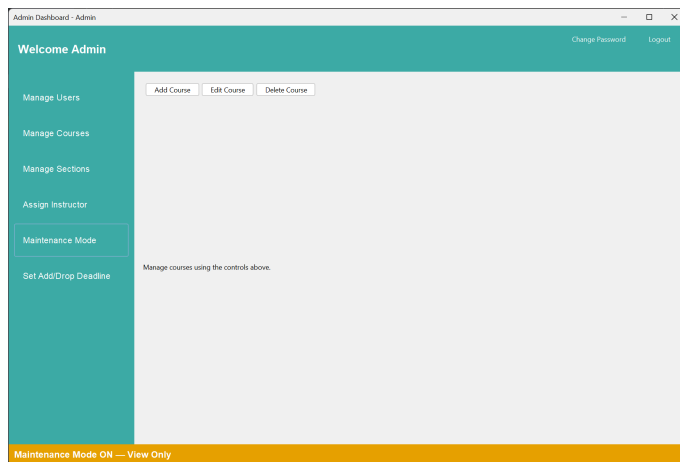
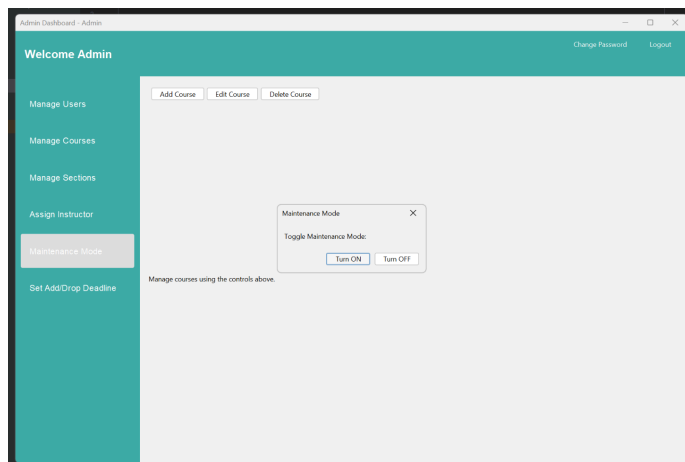
Delete Course with sections and students blocked



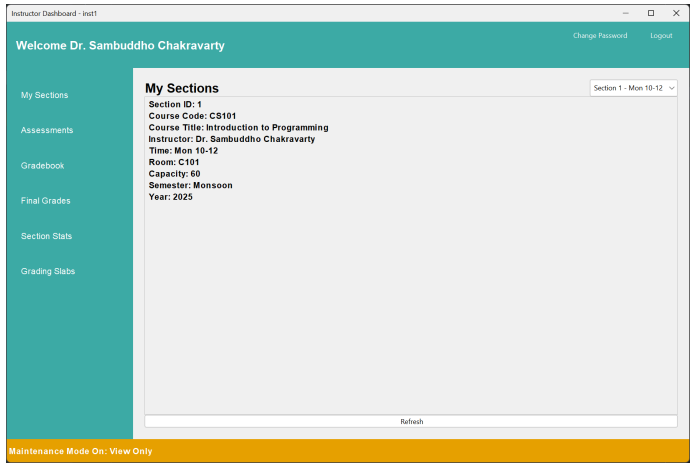
Assign Instructor



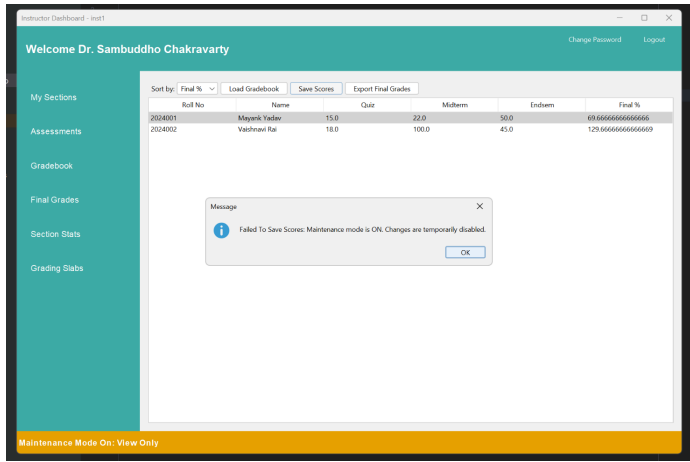
Maintenance Mode Toggle



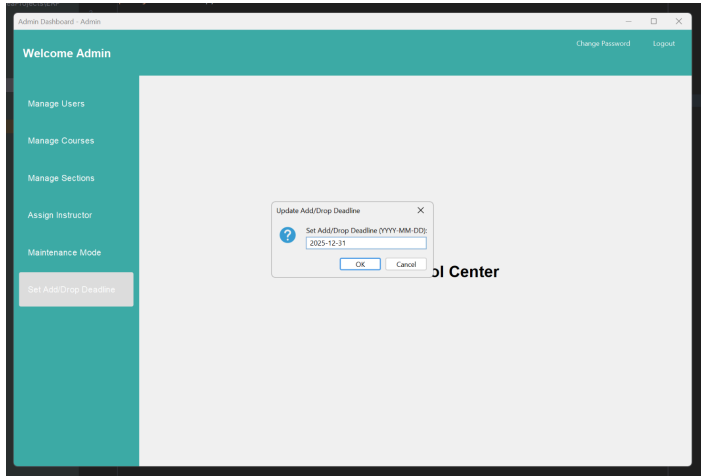
Maintenance Flag



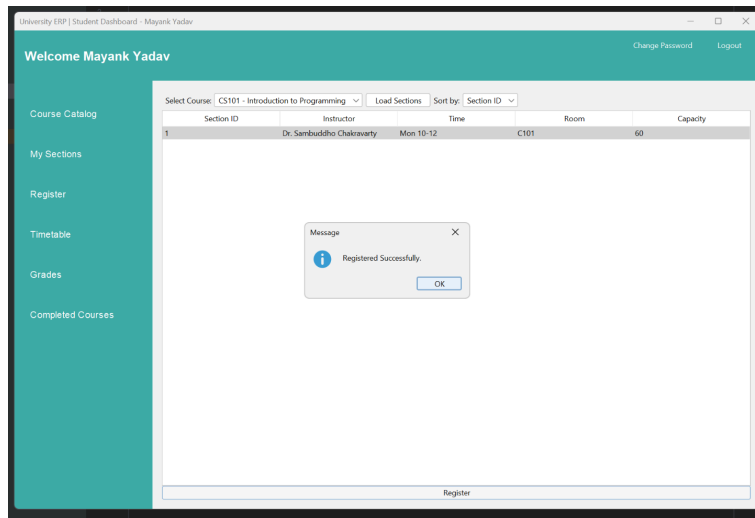
Writes blocked during maintenance



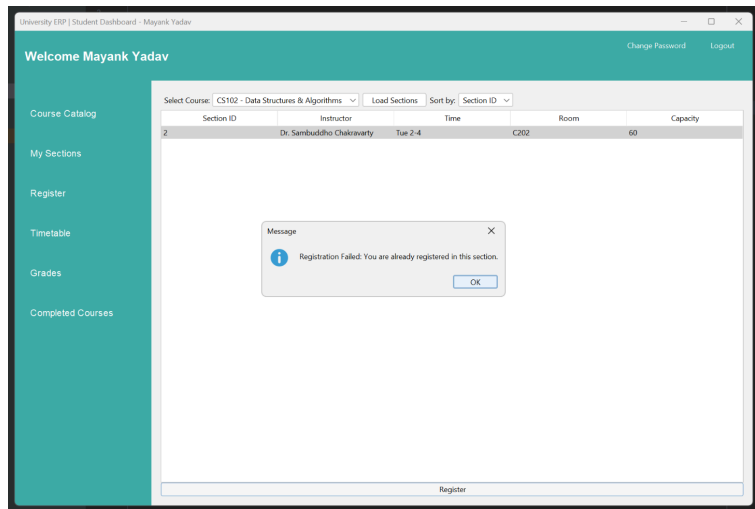
Add Drop deadline setting



Course Registration (After logging in again after changing deadline, and 5 minute lock lifted on stu1)



Duplicate Registration Blocked



Final Grade Weighting Rule

The final percentage for a student in a section is computed by multiplying the score in percentage for each assessment (student score/ max score) with the weight of the assessment in percentage (so 0.9 for 90% etc.).

The letter grade or final grade for a course is calculated based on where the final percentage of a student falls in the slabs defined by the instructor.

Maintenance Mode Implementation

Maintenance mode is implemented by showing a UI Banner at the bottom of the screen whenever maintenance is on.

All functions which write to the database call the function `requireWriteAllowed()` before modifying the database, which throws an exception if the maintenance mode is on, blocking the write. An appropriate message is shown.

Access Management Implementation

After successful login, the current user is saved as a user object. The `AccessManager` class provides functions to check if the current user is an admin, student or instructor. There is also a function to check if the instructor teaches the selected section.

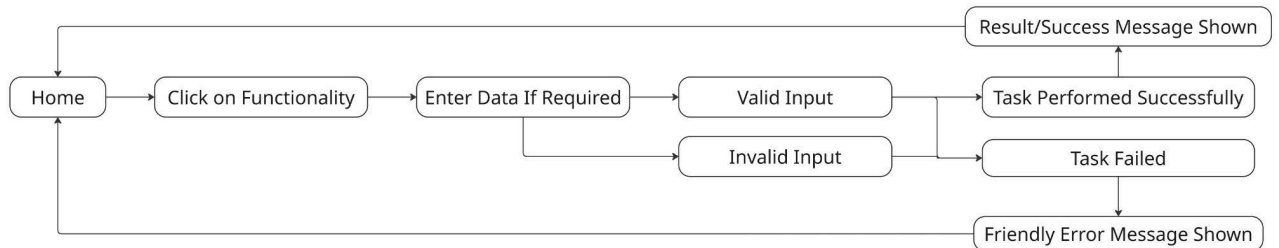
All function calls in the service layer, call the relevant access check before reading/writing/carrying out the functionality, thereby ensuring that the current logged in user and the required user type are matched before anything else happens.

Use-Cases

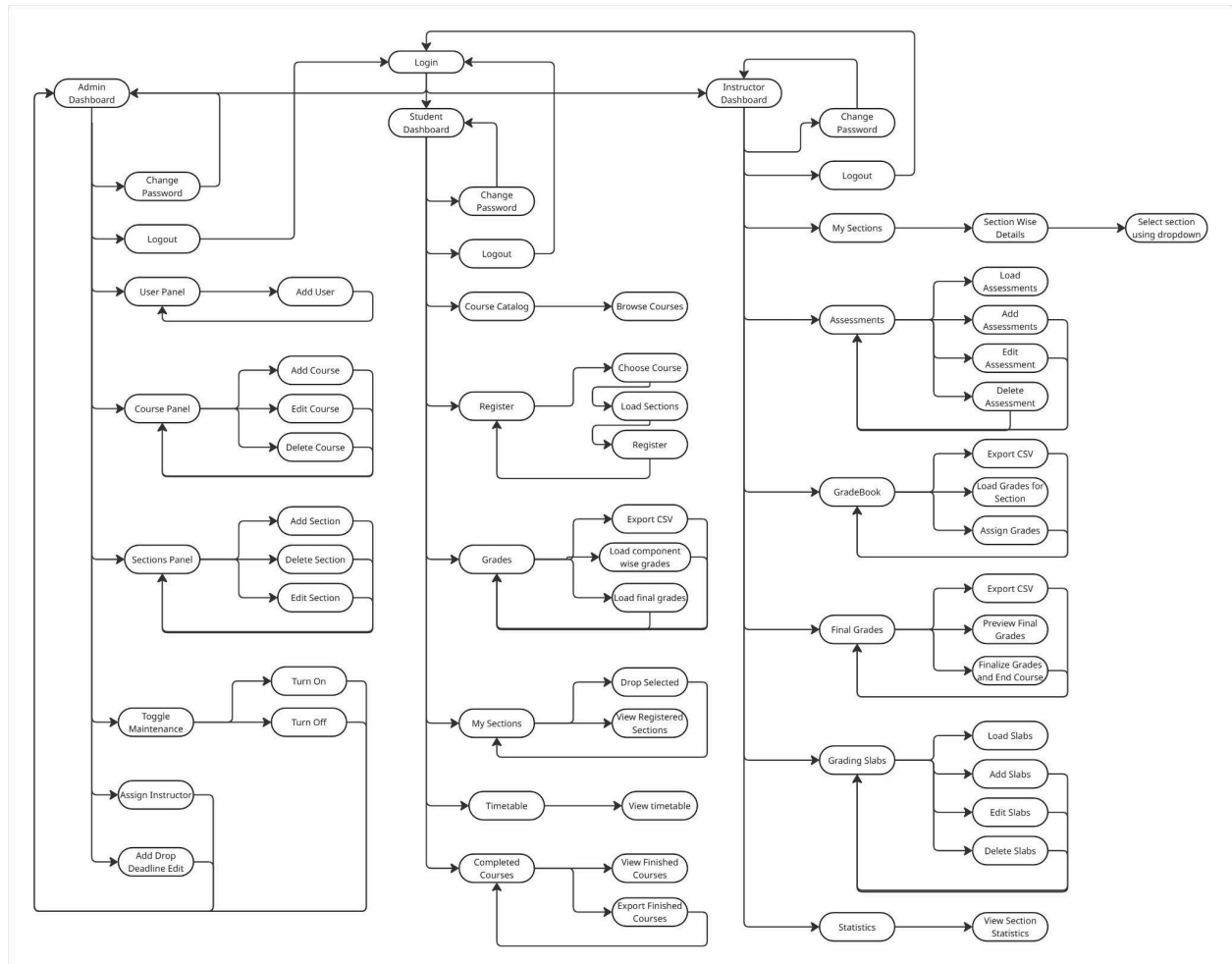
1. Student:
 - a. View course catalog
 - b. View registered sections
 - c. View completed courses
 - d. Register for and drop courses
 - e. View assessment scores and final grades
 - f. Export final grades in a CSV file
2. Instructor:
 - a. View assigned sections
 - b. Create assessments for your section
 - c. Manage assessment scores for your section
 - d. Manage final grade slabs for your section
 - e. View statistics for a section
 - f. Export grades and scores in a CSV file
3. Admin:
 - a. Add users
 - b. Add/Update/Delete courses
 - c. Add/Update/Delete sections

- d. Assign instructor to a section
- e. Toggle maintenance mode
- f. Manage add drop deadline

User Flow For Any Functionality

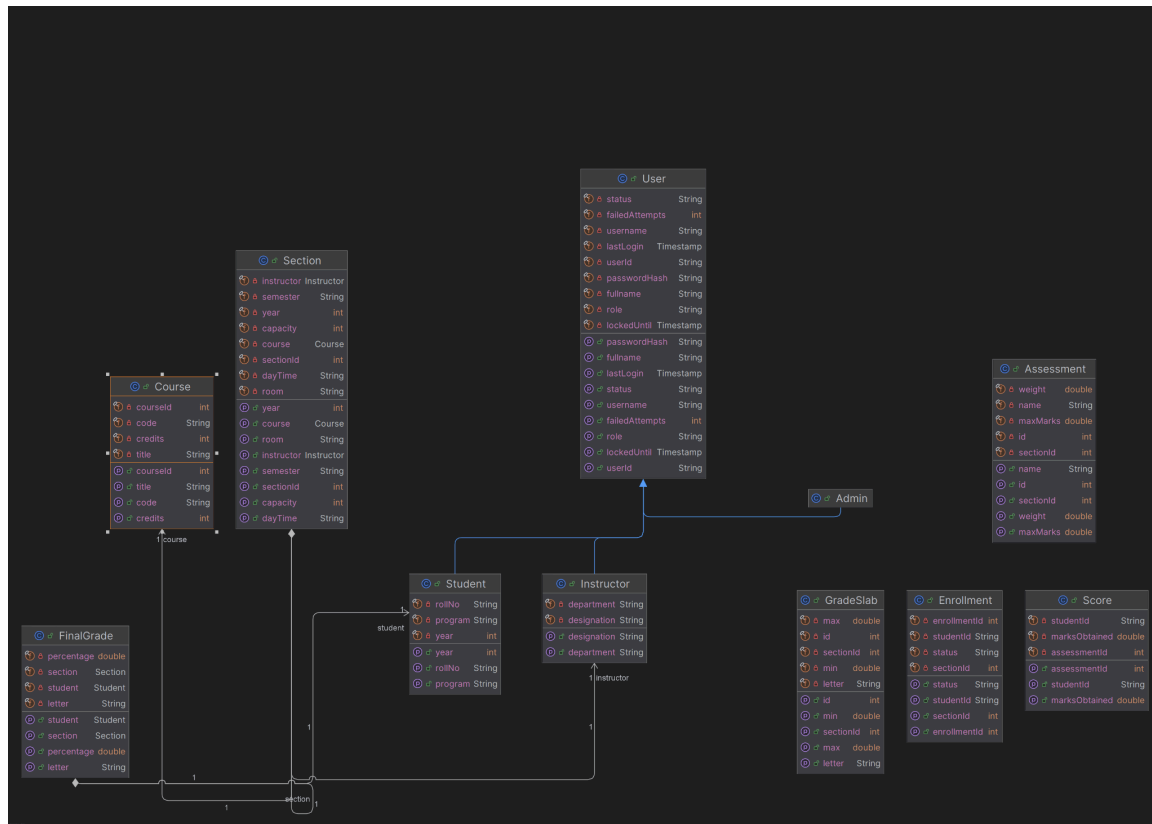


User Flow for Full App

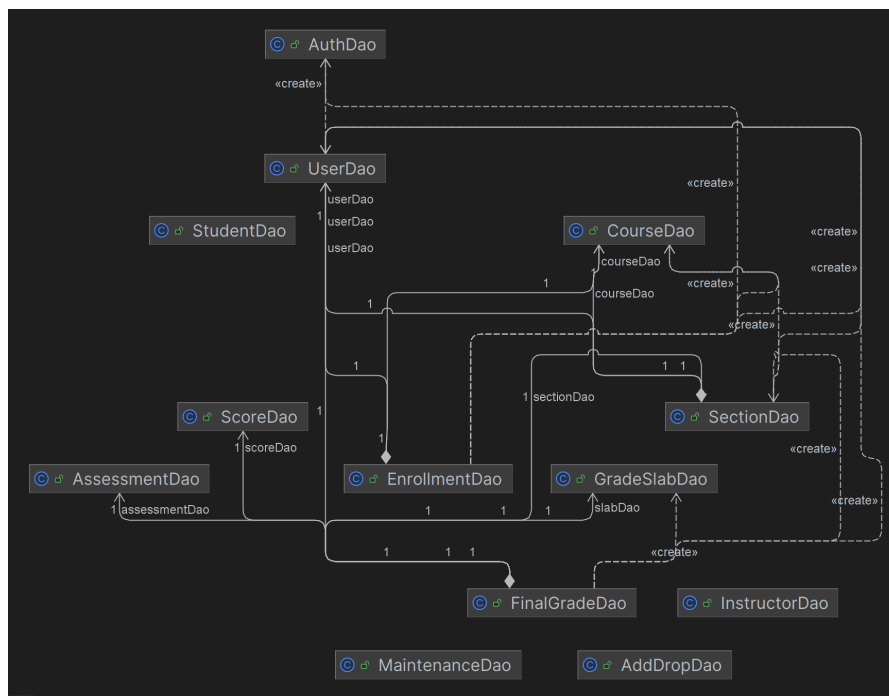


Relationship Diagrams

Domain



Data



Service



API



Access

CurrentSession		
Ⓢ	CurrentSession()	
Ⓜ	get()	User
Ⓜ	set(User)	void
Ⓜ	clear()	void

AccessManager		
Ⓢ	AccessManager(DataSource)	
Ⓜ	current()	User
Ⓜ	requireInstructorForSectionAccess(String, int)	void
Ⓜ	requireStudentAccess(String)	void
Ⓜ	requireInstructorAccess(String)	void
Ⓜ	requireAdminAccess()	void

Auth

PasswordUtil		
Ⓢ	PasswordUtil()	
Ⓜ	hashPassword(String)	String
Ⓜ	main(String[])	void

UI



Inspiration

Overall UI inspired by IIIT Delhi ERP (colours scheme, left nav, top bar with welcome & logout, change password on the top right).